

Prototype-routed exponential-gated residual learning for multi-horizon regional electricity-load forecasting

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ABSTRACT

Short-term electricity-load forecasting (STLF) supports reliable dispatch, reserve scheduling, renewable integration, demand-response planning, and low-carbon power-system operation. Existing forecasting models often struggle to jointly capture heterogeneous load-weather regimes, horizon-dependent residual errors, and causal reliability control for multi-horizon prediction. To address this gap, this study proposes a Prototype-Routed eXponential-Gated Residual Network (PRX-Net) for six-hour-ahead regional electricity-load forecasting. PRX-Net combines a prototype-routed randomized backbone, recurrent residual correction, and causal previous-error gating. The randomized backbone uses global and local enhanced edRVFL experts to model broad and regime-specific load-weather behavior, while the recurrent module estimates the remaining horizon-dependent residual. The exponential gate controls the residual contribution using matured error evidence from previous forecast origins. Experiments on three regional load-weather datasets show that PRX-Net achieves root-mean-square error (RMSE) = 740.82, mean absolute error (MAE) = 528.89, mean absolute percentage error (MAPE) = 2.02%, R^2 = 0.9808, normalized RMSE (NRMSE) = 2.90%, and normalized MAE (NMAE) = 2.09%. Compared with edRVFL, iTransformer, and PatchTST, PRX-Net reduces average RMSE by 12.91%, 2.49%, and 11.69%, respectively. These results show that combining regime-aware randomized forecasting with gated residual correction improves multi-horizon regional load prediction.

1. Introduction

Reliable electricity-load forecasting is becoming a core enabling layer for sustainable power-system operation. The United Nations Sustainable Development Goals emphasize affordable and clean energy through SDG 7 and climate action through SDG 13, while the power sector must simultaneously expand renewable integration, maintain reliability, and operate demand-side flexibility at regional scale [1,2]. These objectives increase the need for accurate short-term load forecasts because many operational decisions depend on anticipated demand rather than observed demand alone. Forecast-informed model predictive control in building energy systems, for example, explicitly depends on load and disturbance predictions to optimize storage and demand-response decisions, and recent studies show that forecast quality directly affects control performance [3,4]. In parallel, microgrid energy-management and control formulations often optimize dispatch under uncertain photovoltaic generation and load trajectories, which further makes demand

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forecasting a prerequisite for low-carbon, reliable, and economically efficient operation [5]. Related work further shows that deep-learning-based net-load forecasting supports power-system flexibility analysis under distributed energy resources and electric-vehicle charging [6]. Recent energy-forecasting and dispatch studies also show that physics-informed learning, edge-aware forecasting, and time-varying economic dispatch benefit from reliable forecast information when the operating environment changes over time [7–10]. Short-term electricity-load forecasting therefore serves not only as a prediction task, but also as an information interface between data-driven demand modeling and downstream sustainable energy-control decisions.

Existing forecasting models for electricity-load and general time-series forecasting can be broadly organized into statistical models, machine-learning models, deep-learning models, and randomized ensemble networks. A recent application-driven review further organizes load-forecasting methods according to how their outputs support downstream operational decisions [11]. Statistical time-series models provide the earliest and most interpretable forecasting layer. Autoregressive (AR) models capture dependence on past observations, while Autoregressive Moving Average (ARMA) models add moving-average error terms to represent serially correlated shocks. Autoregressive Integrated Moving Average (ARIMA) models extend this structure through differencing when nonstationarity appears, and Seasonal ARIMA (SARIMA) further introduces seasonal autoregressive, differencing, and moving-average components for periodic demand cycles. Recent load-forecasting studies still use ARIMA and SARIMA because they provide transparent baselines and stable short-horizon behavior under clear seasonal structure [12,13]. However, these models often depend on linear assumptions, fixed seasonal orders, and limited external-variable interactions, so they can underfit nonlinear load-weather responses and heterogeneous operating regimes.

To address nonlinear feature interactions, machine-learning models introduce flexible mappings between calendar variables, lagged load, and meteorological inputs. Support Vector Regression (SVR) uses kernel functions to map input features into nonlinear regression spaces. Random Forest (RF) reduces variance through bagged decision trees and captures threshold-like interactions among lagged and weather variables. Gradient-boosted tree methods such as XGBoost improve this idea by sequentially correcting residuals of weak learners and have shown strong short-term load-forecasting performance under engineered time, load, and weather features [14,15]. More recent studies further combine feature selection, transfer learning, or tree ensembles to improve performance in data-poor or regional forecasting settings [16,17]. Stacking-based ensemble frameworks that combine decomposition and transfer learning have also been shown to improve robustness and generalization for electric load forecasting [18]. However, these models usually depend on manually constructed lag features and treat the forecasting horizon as a supervised regression target rather than an explicit temporal correction process.

To address sequential dependence more directly, deep-learning models learn temporal representations from ordered input windows. Recent LSTM-based and dynamically optimized recurrent forecasting studies use gated recurrent memory to capture nonlinear temporal persistence in energy time-series prediction [19]. Temporal Convolutional Networks (TCNs) replace recurrence with causal dilated convolutions and provide long effective memory with parallel training [20]. Transformer-based forecasters then use attention mechanisms to model long-range dependencies, and recent time-series variants improve the original design by adapting tokenization and variable representation. PatchTST converts time series into patch tokens and uses channel-independent modeling to reduce attention cost and preserve local temporal structure [21]. iTransformer inverts the tokenization axis so that variates become tokens, allowing attention to model cross-variable dependence more naturally [22]. In parallel, recent energy-forecasting studies show that physics-informed spatio-temporal learning, recurrent architecture optimization, and hardware-aware Transformer-LSTM exploration can improve forecasting reliability under practical deployment constraints [8,19,23]. However, fully trainable deep models can require careful tuning, large data support, and high model capacity, while their global mappings may still miss localized demand regimes or produce residual errors that vary across forecast horizons.

To reduce training burden while preserving nonlinear representation, randomized neural networks provide another model family. Random Vector Functional Link (RVFL) networks generate random hidden features and learn output weights in closed form, which avoids expensive end-to-end backpropagation. Deep RVFL (dRVFL) stacks randomized hidden layers to improve representation depth, and ensemble dRVFL (edRVFL) uses layer-wise outputs as an implicit ensemble [24]. In short-term load forecasting, Gao et al. apply edRVFL with walk-forward empirical wavelet decomposition and show that randomized deep ensembles can provide strong accuracy with efficient training [25]. Later work extends this line toward online dynamic edRVFL for sequential forecasting and stacked residual edRVFL for medium-scale time-series forecasting [26,27]. In related time-series forecasting, mixture-of-experts designs further indicate that adaptive expert specialization can help models separate regular and rare temporal regimes [28]. More directly relevant to the present problem, a cycle-aware adaptive mixture-of-experts architecture with a high-demand-sensitive asymmetric loss has recently been proposed for electrical load forecasting [29], and a regime-conditioned horizon-aware mixture-of-experts design has been proposed for spatio-temporal wind forecasting [30], indicating a broader recent trend toward regime-aware and horizon-aware expert routing in energy time-series forecasting. However, existing edRVFL variants mainly strengthen global randomized representation or online updating; they do not explicitly route load-weather states into local regime experts, and they do not regulate recurrent residual corrections through causal horizon-wise error evidence.

Taken together, existing forecasting models still leave three practical gaps for multi-horizon regional load forecasting. First, global statistical, machine-learning, and deep-learning models can miss heterogeneous load-weather regimes because they usually learn a single mapping across different operating states. Second, deep sequence models can capture temporal dependence, but they often forecast the target directly rather than explicitly correcting the residual left by an efficient backbone. Third, many hybrid and ensemble forecasters combine model outputs without checking whether the residual correction is reliable at each forecast horizon. These limitations motivate a forecasting architecture that combines regime-aware local modeling, residual learning, and causal horizon-wise reliability control.

To address these challenges, this work proposes PRX-Net, a Prototype-Routed eXponential-Gated Residual Network for six-hour-ahead regional electricity-load forecasting. The prototype-routed randomized backbone module addresses regime heterogeneity by routing load-weather descriptors to global and local enhanced edRVFL experts. The recurrent residual correction module addresses horizon-dependent temporal errors by learning the residual around the randomized backbone instead of replacing it. The exponential previous-error gate addresses reliability control by regulating the residual contribution using matured backbone and corrected-candidate errors from previous forecast origins.

The main contributions of this work are summarized as follows:

- A prototype-routed randomized backbone is proposed to combine global and local enhanced edRVFL experts, enabling regime-aware load-weather modeling while retaining efficient randomized output learning.
- A recurrent residual correction module is developed to model horizon-dependent temporal errors around the randomized backbone, so recurrent learning refines the forecast instead of replacing the primary forecasting path.
- A causal exponential previous-error gate is introduced to regulate the residual contribution at each forecast horizon using matured error evidence, improving the reliability of multi-horizon forecast fusion.

The rest of this paper is organized as follows. Section 2 presents the PRX-Net architecture, including load-weather feature construction, prototype-routed randomized forecasting, enhanced edRVFL experts, recurrent residual correction, and causal exponential previous-error gating. Section 3 describes the regional load-weather datasets, chronological forecasting protocol, compared methods, ablation variants, hyperparameter selection, and evaluation metrics. Section 4 reports the ablation study, baseline comparison, horizon-wise analysis, and Wilcoxon–Holm statistical tests. Section 5 concludes the paper and discusses limitations and future research directions.

2. Proposed methodology

This study proposes PRX-Net, a Prototype-Routed eXponential-Gated Residual Network for multi-horizon short-term electricity-load forecasting. At each forecast origin t , the model receives a normalized multivariate historical window $\mathbf{X}_t \in \mathbb{R}^{T \times F}$, where T is the look-back length and F is the number of input variables. The target is the next H -step load vector $\mathbf{y}_t \in \mathbb{R}^H$. PRX-Net predicts this target through three coupled stages. First, a prototype-routed randomized backbone module (PRRBM) maps load-weather descriptors to a backbone forecast using global and prototype-specific enhanced edRVFL experts. Second, a recurrent residual correction module (RRCM) estimates the horizon-dependent temporal error that remains around the randomized backbone. Third, an exponential previous-error gate regulates the residual contribution using matured causal error evidence from previous forecast origins. This layered design keeps randomized learning as the primary forecasting path, uses recurrence only for residual dynamics, and applies horizon-wise reliability control before forming the final forecast.

PRX-Net represents each input window through a load-weather descriptor vector formed from a selectable descriptor pool that includes the autoregressive trajectory and optional statistical, trend, spectral, and calendar summaries. The flattened history preserves the recent multivariate evolution, while the selected additional descriptors expose load variation, short-term slope, periodic structure, or calendar recurrence to the randomized learner.

$$\boldsymbol{\phi}_t = [\text{vec}(\mathbf{X}_t), \boldsymbol{\mu}_t, \boldsymbol{\sigma}_t, \mathbf{s}_t, \mathbf{f}_t, \mathbf{c}_t]. \quad (1)$$

Here, $\boldsymbol{\phi}_t$ is the constructed descriptor vector, $\text{vec}(\mathbf{X}_t)$ is the flattened input window, $\boldsymbol{\mu}_t$ and $\boldsymbol{\sigma}_t$ are mean and standard-deviation descriptors, $\mathbf{s}_t$ contains slope descriptors, $\mathbf{f}_t$ contains frequency-domain descriptors, and $\mathbf{c}_t$ contains cyclic calendar descriptors. The active subset is selected on the validation set. For these candidate descriptor blocks, if $x_{t,i,f}$ denotes the normalized value of feature f at lag i within the look-back window, then $\boldsymbol{\mu}_t$ and $\boldsymbol{\sigma}_t$ are computed feature-wise over the T lags as $\mu_{t,f} = T^{-1} \sum_{i=1}^T x_{t,i,f}$ and $\sigma_{t,f} = \sqrt{T^{-1} \sum_{i=1}^T (x_{t,i,f} - \mu_{t,f})^2}$. The slope descriptor $\mathbf{s}_t$ stores the least-squares linear trend coefficient of each feature over a standardized lag index, so it summarizes short-term increasing or decreasing behavior within the input window. The frequency descriptor $\mathbf{f}_t$ contains the magnitudes of the first nonzero real-FFT components of the dynamic load-weather variables, which gives a compact representation of periodic variation in the recent history. When selected, the calendar descriptor $\mathbf{c}_t$ uses sine-cosine encodings of the calendar variables at the forecast origin, such as hour-of-day and month-of-year, to represent cyclic temporal recurrence without introducing artificial discontinuities. The randomized backbone uses $\boldsymbol{\phi}_t$ to produce the main forecast, whereas the recurrent residual module uses the original sequence $\mathbf{X}_t$ to learn the remaining temporal error.

The complete PRX-Net forecast can therefore be expressed as a gated mapping from the randomized backbone forecast, the recurrent residual estimate, and the matured previous-error evidence:

$$\hat{\mathbf{y}}_t = \mathcal{G}_{\text{exp}}(\hat{\mathbf{y}}_t^B, \hat{\mathbf{r}}_t, \mathbf{E}_{t-1}). \quad (2)$$

In this expression, $\hat{\mathbf{y}}_t \in \mathbb{R}^H$ is the final H -step forecast, $\hat{\mathbf{y}}_t^B \in \mathbb{R}^H$ is the PRRBM backbone forecast, and $\hat{\mathbf{r}}_t \in \mathbb{R}^H$ is the RRCM residual estimate. The term $\mathbf{E}_{t-1}$ represents matured previous-error evidence available before issuing the forecast at origin t , and $\mathcal{G}_{\text{exp}}(\cdot)$ denotes the exponential gating operator that assigns horizon-wise trust to the residual correction. This compact form gives the overall forecasting path of PRX-Net. The following subsections define the prototype-routed randomized backbone, enhanced edRVFL expert, recurrent residual module, and exponential previous-error gate in detail.

Fig. 1 illustrates the block architecture of PRX-Net. The descriptor builder forms $\boldsymbol{\phi}_t$ from the input window, prototype routing assigns soft regime weights to the global and local edRVFL experts, and the PRRBM backbone combines these expert forecasts. In

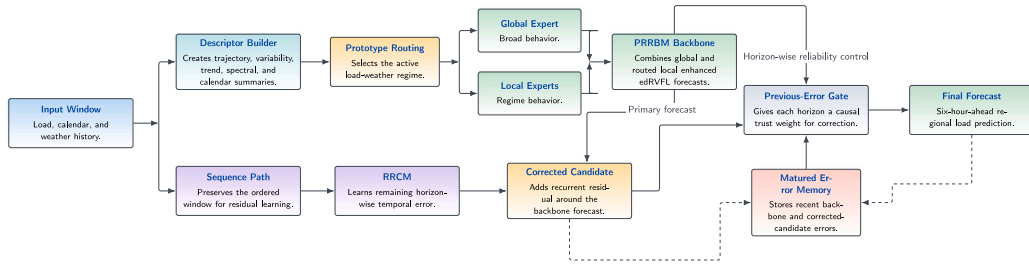


Fig. 1. Block architecture of PRX-Net, showing the prototype-routed randomized backbone module (PRRBM), the recurrent residual correction module (RRCM), and the exponential previous-error gate.

parallel, the sequence path preserves the ordered input window for the RRCM, which estimates the horizon-dependent residual and forms the residual-corrected candidate around the backbone forecast. The exponential previous-error gate then uses the matured backbone and corrected-candidate error memory to assign a causal, horizon-wise trust weight to the residual correction and produce the final forecast.

2.1. Prototype-routed randomized backbone module

To address the weak adaptation of a single global randomized mapping to heterogeneous load-weather regimes, PRX-Net uses the prototype-routed randomized backbone module. PRRBM learns a set of prototypes in the constructed feature space and assigns each input window to these prototypes through soft routing weights. These prototypes represent local operating regimes, while a global expert remains active to stabilize the prediction when the input window does not belong strongly to one regime. The module therefore keeps the efficiency of randomized learning while allowing the backbone to adapt to heterogeneous demand states.

$$q_{t,m} = \frac{\exp(-d(\phi_t, \mathbf{p}_m)/\tau)}{\sum_{j=1}^M \exp(-d(\phi_t, \mathbf{p}_j)/\tau)}. \quad (3)$$

where $q_{t,m}$ denotes the routing weight of sample t for prototype m , ϕ_t denotes the constructed feature vector, $\mathbf{p}_m$ denotes the m th prototype, $d(\cdot, \cdot)$ denotes the distance function, τ denotes the routing temperature, and M denotes the number of prototypes. In this study, $d(\phi_t, \mathbf{p}_m)$ is the squared Euclidean distance, $d(\phi_t, \mathbf{p}_m) = \|\phi_t - \mathbf{p}_m\|_2^2$, computed in the standardized descriptor space.

The backbone forecast combines the global randomized expert and the prototype-routed local experts. The global expert captures broad load dynamics, while the local experts focus on prototype-specific behavior.

$$\hat{\mathbf{y}}_t^B = \lambda_g \hat{\mathbf{y}}_t^G + (1 - \lambda_g) \sum_{m=1}^M q_{t,m} \hat{\mathbf{y}}_t^m. \quad (4)$$

where $\hat{\mathbf{y}}_t^B$ denotes the PRRBM backbone forecast, $\hat{\mathbf{y}}_t^G$ denotes the global expert forecast, $\hat{\mathbf{y}}_t^m$ denotes the forecast from local expert m , $q_{t,m}$ denotes the corresponding routing weight, and λ_g denotes the global-expert mixing coefficient. The coefficient satisfies $\lambda_g \in [0, 1]$, so Eq. (4) forms a convex mixture between the global expert forecast and the prototype-weighted local expert forecast. PRRBM improves regime sensitivity, but each routed expert still needs a nonlinear representation that remains expressive without requiring expensive end-to-end neural training.

2.2. Enhanced edRVFL expert with RSC and BF

Each PRRBM expert therefore uses an enhanced deep random vector functional link model with random skip connections (RSC) and a boosting factor (BF). The expert assigns hidden-layer weights randomly and learns output weights through ridge regression. This design avoids full backpropagation inside the randomized backbone and still produces a deep nonlinear representation. RSC reuses randomly selected previous hidden states so that deeper layers receive both direct input information and earlier randomized transformations. BF controls the strength of the randomized features and improves the contribution of deeper representations.

$$\mathbf{A}_\ell = [\mathbf{A}_1, \mathbf{H}_{\ell-1}, \mathbf{H}_{j_1}, \dots, \mathbf{H}_{j_K}]. \quad (5)$$

where $\mathbf{A}_\ell$ denotes the input matrix of layer ℓ , $\mathbf{A}_1$ denotes the direct-link input matrix, $\mathbf{H}_{\ell-1}$ denotes the immediately preceding hidden representation, $\mathbf{H}_{j_1}, \dots, \mathbf{H}_{j_K}$ denote randomly selected earlier hidden representations, and K denotes the number of random skip states. For layer $\ell > 2$, the eligible random-skip set is defined as $S_\ell = \{1, 2, \dots, \ell-2\}$, which excludes the immediately preceding layer because $\mathbf{H}_{\ell-1}$ is already included in Eq. (5). The skip indices $j_1, j_2, \dots, j_K$ are sampled uniformly without replacement from S_ℓ , with $K \leq |S_\ell|$. Equivalently, a selected skip index can be written as $j = \ell - 1 - k$, where $k > 1$ and $j \in S_\ell$.

$$\mathbf{H}_\ell = g(b_\ell \mathbf{A}_\ell \mathbf{W}_\ell). \quad (6)$$

where $\mathbf{H}_\ell$ denotes the hidden representation at layer ℓ , $g(\cdot)$ denotes the activation function, b_ℓ denotes the layer-wise boosting factor, $\mathbf{A}_\ell$ denotes the layer input matrix, and $\mathbf{W}_\ell$ denotes the random hidden-weight matrix. In the final PRX-Net configuration, $g(\cdot)$ is implemented as the GELU activation function, using $\text{GELU}(z) = 0.5z\{1 + \tanh[\sqrt{2/\pi}(z + 0.044715z^3)]\}$. For each layer, $\mathbf{W}_\ell \in \mathbb{R}^{d_\ell \times D_h}$ is sampled once from a zero-mean Gaussian distribution, $\mathbf{W}_{\ell,ab} \sim \mathcal{N}(0, \sigma_{W,\ell}^2)$, where $\sigma_{W,\ell} = s_{\text{in}}/\sqrt{d_\ell}$, d_ℓ is the layer-input dimension, D_h is the hidden dimension, and s_{in} is the input-scale hyperparameter. The sampled weights are then fixed, and only the output weights are estimated by ridge regression.

Each layer solves its own output mapping, so the expert does not depend on a single terminal hidden layer. The output weight matrix is obtained through ridge regression.

$$\beta_\ell = (\mathbf{D}_\ell^\top \mathbf{D}_\ell + \eta \mathbf{I})^{-1} \mathbf{D}_\ell^\top \mathbf{Y}_{\text{tr}}. \quad (7)$$

where β_ℓ denotes the output weight matrix at layer ℓ , $\mathbf{D}_\ell$ denotes the design matrix formed from direct and hidden features, η denotes the ridge coefficient, $\mathbf{I}$ denotes the identity matrix, and $\mathbf{Y}_{\text{tr}}$ denotes the training target matrix. At prediction time, the same layer construction gives the test design vector $\mathbf{d}_{t,\ell}^m$ for expert m , and the layer-wise forecast used in Eq. (8) is computed as $\hat{\mathbf{y}}_{t,\ell}^m = \mathbf{d}_{t,\ell}^m \beta_\ell^m$.

$$\hat{\mathbf{y}}_t^m = \text{Agg}(\hat{\mathbf{y}}_{t,1}^m, \hat{\mathbf{y}}_{t,2}^m, \dots, \hat{\mathbf{y}}_{t,L}^m). \quad (8)$$

where $\hat{\mathbf{y}}_t^m$ denotes the forecast of expert m , $\hat{\mathbf{y}}_{t,\ell}^m$ denotes the layer-wise forecast from layer ℓ , L denotes the number of randomized layers, and $\text{Agg}(\cdot)$ denotes the aggregation operator. In this study, $\text{Agg}(\cdot)$ is the element-wise median across the L layer-wise forecasts, so each forecast horizon is aggregated independently over the randomized layers. The enhanced expert strengthens the nonlinear randomized backbone, but structured temporal errors can still remain after the backbone forecast, especially when the future trajectory changes across the forecasting horizon.

2.3. Recurrent residual correction module

PRX-Net next introduces the recurrent residual correction module to model the temporal error that remains after the randomized backbone forecast. RRCM learns the residual left by the PRRBM backbone rather than forecasting the full load sequence again. This residual formulation keeps the randomized backbone as the primary forecaster and assigns the recurrent model to the remaining horizon-dependent temporal error. The LSTM receives the original normalized sequence, which allows it to learn sequential correction patterns that are not fully captured by compact randomized features.

$$\mathbf{r}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t^B. \quad (9)$$

where $\mathbf{r}_t \in \mathbb{R}^H$ denotes the backbone residual vector, $\mathbf{y}_t \in \mathbb{R}^H$ denotes the observed target vector, and $\hat{\mathbf{y}}_t^B \in \mathbb{R}^H$ denotes the PRRBM backbone forecast. During training, $\mathbf{r}_t$ is used as the supervised target for the RRCM, so the recurrent module learns to approximate the residual mapping from $\mathbf{X}_t$ to the remaining backbone error.

$$\hat{\mathbf{r}}_t = \mathcal{F}_{\text{RRCM}}(\mathbf{X}_t). \quad (10)$$

where $\hat{\mathbf{r}}_t$ denotes the estimated residual vector, $\mathcal{F}_{\text{RRCM}}(\cdot)$ denotes the recurrent residual correction function, and $\mathbf{X}_t$ denotes the normalized input sequence.

The residual estimate forms a corrected candidate around the backbone forecast. This candidate lets PRX-Net use recurrent information without replacing the backbone structure.

$$\hat{\mathbf{y}}_t^C = \hat{\mathbf{y}}_t^B + (1 + \delta_r) \hat{\mathbf{r}}_t. \quad (11)$$

where $\hat{\mathbf{y}}_t^C$ denotes the residual-corrected candidate forecast, $\hat{\mathbf{y}}_t^B$ denotes the PRRBM backbone forecast, $\hat{\mathbf{r}}_t$ denotes the RRCM residual estimate, and $\delta_r \in [0, 1]$ denotes the residual adjustment margin. Thus, $1 + \delta_r$ calibrates the residual magnitude, while the bounded gate introduced below controls the horizon-wise trust in the corrected residual path. RRCM improves temporal correction, but the corrected candidate does not always remain more reliable than the backbone across all horizons and operating conditions.

2.4. Exponential previous-error gate

PRX-Net finally introduces an exponential previous-error gate to regulate when the recurrent residual pathway should influence the final forecast. The gate maintains separate matured error memories for the backbone forecast and the residual-corrected candidate. It then assigns a horizon-wise trust weight to the recurrent correction. When the corrected candidate produces lower recent error, the gate increases the residual contribution. When the backbone remains more reliable, the gate suppresses the correction. This update uses only matured errors, so the inference path remains causal.

$$\gamma_{t,h} = \frac{\exp(-\beta E_{t-1,h}^C)}{\exp(-\beta E_{t-1,h}^C) + \exp(-\beta E_{t-1,h}^B)}. \quad (12)$$

where $\gamma_{t,h}$ denotes the residual trust weight at horizon h , β denotes the gate sharpness parameter, $E_{t-1,h}^C$ denotes the matured corrected-candidate error memory available before issuing the forecast at time t , and $E_{t-1,h}^B$ denotes the matured backbone error memory available before issuing the forecast at time t .

The final PRX-Net forecast applies this trust weight to the recurrent correction. The model therefore keeps the backbone prediction when the residual pathway becomes uncertain and increases the correction when recent evidence supports it.

$$\hat{y}_{t,h} = \hat{y}_{t,h}^B + \gamma_{t,h}(1 + \delta_r)\hat{r}_{t,h}. \quad (13)$$

where $\hat{y}_{t,h}$ denotes the final PRX-Net forecast at horizon h , $\hat{y}_{t,h}^B$ denotes the PRRBM backbone forecast, $\gamma_{t,h}$ denotes the exponential previous-error gate weight, δ_r denotes the residual adjustment margin, and $\hat{r}_{t,h}$ denotes the RRCM residual estimate. After the corresponding target becomes available, PRX-Net updates the two error memories.

The exponential previous-error gate also provides a simple theoretical reliability control. Because $E_{t-1,h}^B \geq 0$, $E_{t-1,h}^C \geq 0$, and $\beta > 0$, both exponential terms in the gate equation are positive, which gives $0 < \gamma_{t,h} < 1$. Therefore, the residual pathway in the final forecast equation is explicitly moderated by a bounded horizon-wise gate. The gate is also causal because it depends only on $E_{t-1,h}^B$ and $E_{t-1,h}^C$, which are computed from matured previous errors before the forecast at origin t is issued. Moreover, if the corrected candidate has larger matured error than the backbone, $E_{t-1,h}^C > E_{t-1,h}^B$, then $\gamma_{t,h} < 0.5$, and the residual correction is suppressed. If $E_{t-1,h}^C < E_{t-1,h}^B$, then $\gamma_{t,h} > 0.5$, and the residual correction receives more trust. This monotonic relation links the final forecast reliability to previous horizon-wise error evidence rather than to a fixed fusion weight.

$$E_{t,h}^B = \rho E_{t-1,h}^B + (1 - \rho) |y_{t,h} - \hat{y}_{t,h}^B|^\kappa. \quad (14)$$

where $E_{t,h}^B$ denotes the updated backbone error memory at horizon h , ρ denotes the exponential decay coefficient, $y_{t,h}$ denotes the observed load at horizon h , $\hat{y}_{t,h}^B$ denotes the backbone forecast at horizon h , and κ denotes the error-power parameter.

$$E_{t,h}^C = \rho E_{t-1,h}^C + (1 - \rho) |y_{t,h} - \hat{y}_{t,h}^C|^\kappa. \quad (15)$$

where $E_{t,h}^C$ denotes the updated corrected-candidate error memory at horizon h , ρ denotes the exponential decay coefficient, $y_{t,h}$ denotes the observed load at horizon h , $\hat{y}_{t,h}^C$ denotes the residual-corrected candidate forecast at horizon h , and κ denotes the error-power parameter. Together, the proposed architecture moves from feature construction to prototype-routed randomized forecasting, from randomized forecasting to recurrent residual correction, and from residual correction to causal error-gated fusion. The overall pseudocode of the proposed PRX-Net algorithm is given as Algorithm S1 in the Supplementary Material.

Compared with existing methods, the methodological contribution of PRX-Net lies in the coordinated coupling of three functions that are usually handled separately, namely regime-aware randomized forecasting, residual-only recurrent correction, and causal reliability-gated fusion. This design addresses three limitations identified in the preceding methodology discussion, including weak adaptation to heterogeneous load-weather regimes, limited treatment of horizon-dependent residual errors, and fixed or non-causal forecast fusion. In contrast to conventional edRVFL models, PRX-Net does not rely on a single global randomized mapping. Prototype routing separates heterogeneous load-weather states and allows local experts to specialize in regime-specific behavior. In contrast to mixture-of-experts models that mainly route or combine complete expert forecasts, PRX-Net uses routing to form the randomized backbone and then applies recurrent correction only to the remaining forecast residual. In contrast to LSTM, TCN, and Transformer-style forecasters that learn fully trainable global mappings, PRX-Net keeps randomized learning as the primary forecasting path and assigns recurrence to the unresolved temporal error. In contrast to ordinary hybrid, residual-correction, or dynamic ensemble fusion methods, the exponential previous-error gate avoids a fixed or non-causal combination rule and regulates the residual contribution at each horizon using matured backbone and corrected-candidate error evidence. Therefore, the novelty is not the isolated use of routing, residual learning, or gating, but their role-specific integration within a single multi-horizon regional load-forecasting architecture.

3. Experimental setup

The experimental methodology proceeds through five stages, summarized in Fig. 2. Regional load-weather records are aligned into supervised input-target windows and split chronologically into training, validation, and test subsets. PRX-Net is trained on the training subset, while TPE Bayesian optimization with gate-grid search selects hyperparameters on the validation subset. The selected configuration is then applied causally to the held-out test subset, where the compared methods and ablation variants are evaluated using Wilcoxon–Holm significance testing.

3.1. Datasets and forecasting protocol

This study evaluates PRX-Net on three regional electricity-load datasets, TEN (Tennessee), CENT (Central), and SE (Southeast), built from hourly EIA Open Data load records paired with representative NSRDB meteorological records [31–34]. Each dataset contains 43,848 hourly records from 2020-01-01 to 2024-12-31, with $F = 11$ input variables (calendar, load, and weather) and load demand (MW) as the target. Table 1 summarizes the three datasets, including the chronological train/validation/test split, which is identical across regions and gives 7259 supervised windows per region (21,777 in total) after transforming each series into $T = 96$ -hour input, $H = 6$ -hour target samples. Full per-region window counts, target hours, and descriptive load statistics are reported in Tables S3–S4 of the Supplementary Material.

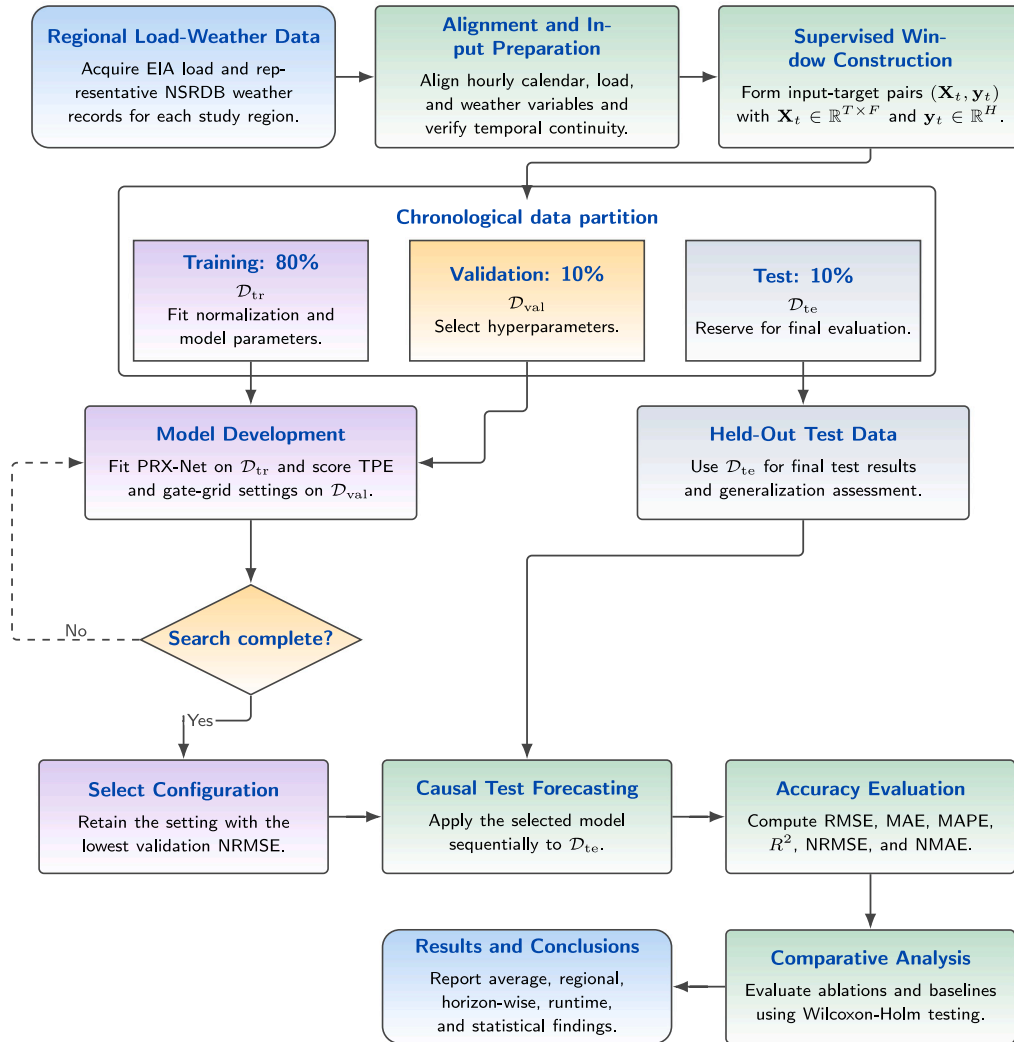


Fig. 2. Flowchart of the experimental methodology, from data acquisition and supervised window construction through chronological partitioning, hyperparameter search, causal test forecasting, and comparative evaluation.

3.2. Ablation variants

The ablation study evaluates the contribution of each PRX-Net component through a progressive model ladder. The ladder starts from the base edRVFL backbone and then adds selected descriptors, prototype routing, separate and joint RSC/BF backbone variants, recurrent residual correction, and exponential previous-error gating. This design isolates the role of each module and shows how the final architecture improves through cumulative component integration. Table 2 lists the ablation variants and the component contribution tested by each variant.

3.3. Hyperparameter optimization and implementation

The validation subset guides hyperparameter selection for all methods. For PRX-Net, TPE Bayesian optimization selects the shared PRRBM/RRCM configuration by minimizing mean validation NRMSE over TEN, CENT, and SE, while dataset-wise grid search selects the exponential previous-error gate per region; selected values and search ranges are reported in Table 3, where N_{LSTM} and U_{LSTM} denote the number of LSTM layers and hidden units in the RRCM, and the three optimal values reported for ρ and κ correspond to TEN, CENT, and SE, respectively. The neural baselines are tuned under the same validation protocol, and each method reports final performance on the held-out test subset.

The selected configuration is compact and regularized ($M = 3$ routing regimes, moderate backbone depth, strong ridge regularization), and the region-specific gate memory tracks relative load variability (Table S4): longest for TEN (highest CV),

Table 1

Overview of the three regional load-weather datasets, including the chronological train/validation/test split (identical across regions).

Code	EIA load region	NSRDB location	Latitude	Longitude	Raw hours	Period	$T \rightarrow H$	Train (%)	Val (%)	Test (%)
TEN	Tennessee	Tennessee	35.5175	-86.5804	43,848	2020-01-01 to 2024-12-31	96 h $\rightarrow$ 6 h	80.31	9.84	9.85
CENT	Central	Kansas	39.5000	-98.3500	43,848	2020-01-01 to 2024-12-31	96 h $\rightarrow$ 6 h	80.31	9.84	9.85
SE	Southeast	Georgia	33.0000	-84.0000	43,848	2020-01-01 to 2024-12-31	96 h $\rightarrow$ 6 h	80.31	9.84	9.85

Table 2

Ablation variants used to evaluate the contribution of each PRX-Net component.

Variant	Model	Purpose	Variant	Model	Purpose
A1	edRVFL backbone	Base randomized forecasting block	A2	edRVFL with selected descriptors	Effect of statistical, slope, spectral, and calendar features
A3	Prototype-routed edRVFL	Effect of regime-aware prototype routing	A4.1	PRRBM with RSC	Separate effect of random skip connections
A4.2	PRRBM with BF	Separate effect of randomized-feature scaling	A4.3	PRRBM with RSC and BF	Joint effect of enhanced randomized representation
A5	PRRBM with RRCM	Effect of recurrent residual correction	A6	PRX-Net	Effect of exponential previous-error gating

Table 3

PRX-Net hyperparameters, optimal ranges, and optimal values.

Parameter	D_ϕ	M	τ	λ_g
Range	Flags, $K_{\text{FFT}} \in \{0, 4, 8, 12, 16, 20\}$	$[2, 7]$	$[0.2, 20]$, log	$[0, 1]$
Optimal	Flat, std, slope, $K_{\text{FFT}} = 16$	3	1.00	0.66
Parameter	L	D_h	η	$g(\cdot)$
Range	$[3, 12]$	$\{64, \dots, 512\}$	$[10^{-2}, 10^2]$, log	{sigmoid,tanh,ReLU,Tribas,GELU}
Optimal	5	384	58.97	GELU
Parameter	s_{in}	BF	N_{LSTM}	U_{LSTM}
Range	$[0.05, 2.0]$, log	$[0.05, 3.0]$, log	$[1, 3]$	$\{16, 32, 64, 128, 256\}$
Optimal	0.20	1.50	1	128
Parameter	δ_r	β	ρ	κ
Range	$\{0, 0.25, 0.50\}$	$\{0.5, 1, 2, 4, 8, 12, 16, 20\}$	$\{0.80, \dots, 0.98\}$	$\{1.0, \dots, 3.0\}$
Optimal	0.25	12	0.90/0.80/0.85	2.00/1.00/1.00

shortest for CENT (lowest CV). Validation sensitivity of the exponential previous-error gate parameters is reported in Table S2, and extended per-parameter interpretation is provided in the Supplementary Material. All methods train on the same processed tensors and forecasting protocol, so performance differences reflect model design rather than data handling.

3.4. Evaluation protocol

All models use the same input length, forecast horizon, chronological data split, target definition, and normalization procedure, so reported differences reflect model behavior rather than evaluation setting. Forecasting accuracy is reported using RMSE, MAE, MAPE, R^2 , NRMSE, and NMAE:

$$\text{RMSE} = \sqrt{\frac{1}{N_s H} \sum_{t=1}^{N_s} \sum_{h=1}^H (y_{t,h} - \hat{y}_{t,h})^2}, \quad \text{MAE} = \frac{1}{N_s H} \sum_{t=1}^{N_s} \sum_{h=1}^H |y_{t,h} - \hat{y}_{t,h}|. \quad (16)$$

$$\text{MAPE} = \frac{100}{N_s H} \sum_{t=1}^{N_s} \sum_{h=1}^H \left| \frac{y_{t,h} - \hat{y}_{t,h}}{y_{t,h}} \right|, \quad R^2 = 1 - \frac{\sum_{t=1}^{N_s} \sum_{h=1}^H (y_{t,h} - \hat{y}_{t,h})^2}{\sum_{t=1}^{N_s} \sum_{h=1}^H (y_{t,h} - \bar{y}_s)^2}. \quad (17)$$

$$\text{NRMSE} = 100 \times \frac{\text{RMSE}}{\bar{y}_{\text{tr}}}, \quad \text{NMAE} = 100 \times \frac{\text{MAE}}{\bar{y}_{\text{tr}}}. \quad (18)$$

where N_s denotes the number of samples in the evaluated split s , H denotes the forecast horizon, $y_{t,h}$ and $\hat{y}_{t,h}$ denote the observed and predicted load, and $\bar{y}_s$ and $\bar{y}_{\text{tr}}$ denote the mean observed load in split s and in the training subset, respectively.

RMSE and MAE report absolute error in the original load scale; MAPE reports percentage error; R^2 reports the fraction of load variance explained by the forecast; and NRMSE and NMAE express RMSE and MAE relative to the training-set mean load, giving a fairer comparison across the differently scaled TEN, CENT, and SE regions.

The statistical analysis evaluates whether the observed accuracy differences persist across paired evaluation units. The test compares PRX-Net with each baseline using paired errors computed across datasets and forecast horizons. This work uses the Wilcoxon signed-rank test because it does not require a normality assumption for the paired differences. When several baselines are

Table 4
Cross-dataset average ablation performance across TEN, CENT, and SE.

Model	RMSE ↓	MAE ↓	MAPE (%) ↓	R^2 ↑	NRMSE (%) ↓	NMAE (%) ↓
A6	740.82	528.89	2.02	0.9808	2.90	2.09
A5	750.30	535.22	2.04	0.9803	2.94	2.12
A4.3	790.05	564.86	2.16	0.9779	3.13	2.25
A4.1	791.68	565.73	2.17	0.9778	3.13	2.25
A4.2	793.49	566.80	2.17	0.9777	3.14	2.26
A3	813.15	578.31	2.21	0.9768	3.21	2.30
A2	847.08	597.34	2.29	0.9744	3.37	2.39
A1	850.65	599.05	2.29	0.9742	3.38	2.40

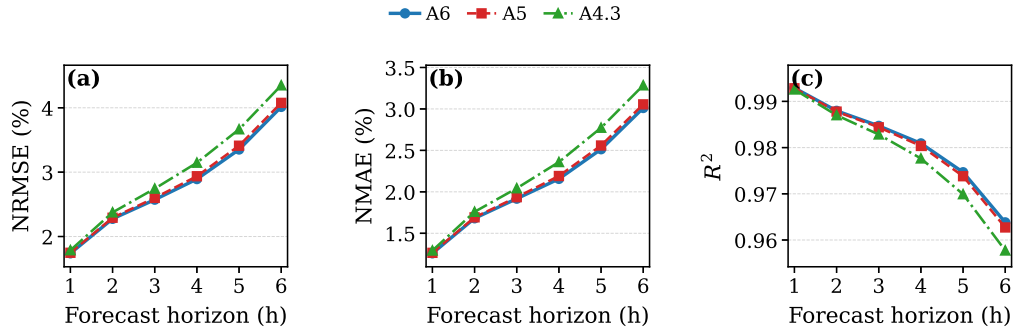


Fig. 3. Cross-dataset horizon-wise NRMSE, NMAE, and R^2 comparison of selected ablation variants.

tested against PRX-Net, Holm correction controls the family-wise error rate. This procedure supports the average metric comparisons with a paired nonparametric significance test.

4. Results and comparison

This section evaluates PRX-Net from three complementary perspectives. First, the ablation study quantifies how each architectural component contributes to forecasting accuracy. Second, the comparison with representative forecasting baselines evaluates the final model against randomized, recurrent, convolutional, and Transformer-style methods. Third, the Wilcoxon–Holm analysis tests whether the observed gains persist across paired dataset-horizon evaluation units. All experiments use the same chronological test subsets of TEN, CENT, and SE. The tables report RMSE, MAE, MAPE, R^2 , NRMSE, and NMAE for all evaluated models, while the horizon-wise figures show selected high-ranking models from the corresponding comparison group. In each horizon-wise figure, panel (a) reports NRMSE, panel (b) reports NMAE, and panel (c) reports R^2 .

4.1. Ablation study

Table 4 reports the cross-dataset average of the ablation variants. Within the A4 group, A4.3 (joint RSC and BF) gives the strongest PRRBM-only result, improving over A3 from RMSE 813.15 to 790.05, MAE 578.31 to 564.86, and MAPE 2.21% to 2.16%. The full PRX-Net model (A6) achieves the best overall performance, with RMSE = 740.82, MAE = 528.89, MAPE = 2.02%, R^2 = 0.9808, NRMSE = 2.90%, and NMAE = 2.09%, reducing average RMSE, MAE, and MAPE by 12.91%, 11.71%, and 12.08% over the base edRVFL backbone, and by a further 1.26%, 1.18%, and 1.20% over A5 PRRBM-RRCM. These gains confirm that the improvement comes from the cumulative integration of prototype routing, enhanced randomized representation, recurrent residual correction, and causal error-gated fusion.

Per-region ablation results follow the same pattern as the cross-dataset average. PRX-Net (A6) ranks first across all metrics on Tennessee (RMSE = 517.21, MAPE = 2.07%), Central (RMSE = 947.85, MAPE = 1.89%), and Southeast (RMSE = 757.40, MAPE = 2.08%). The gain over the edRVFL backbone is largest on Southeast (14.13% RMSE reduction) and smallest on Central (6.67%), where the recurrent residual correction already captures most of the remaining temporal error. Full per-region tables and horizon-wise figures for Tennessee, Central, and Southeast are reported in Tables S5–S7 and Figures S1–S3 of the Supplementary Material.

Fig. 3 summarizes the selected high-ranking ablation variants after averaging horizon-wise results across the three datasets. The figure confirms the table-level trend. A6 PRX-Net gives the lowest normalized errors and the highest R^2 , followed by A5 PRRBM-RRCM and A4.3 PRRBM-RSC-BF. This ordering supports the proposed model ladder because the strongest variants combine prototype routing, enhanced randomized representation, and residual correction, while the final error gate gives the best aggregate accuracy.

Table S1 of the Supplementary Material provides an additional comparison of Naive, ARIMA, SARIMA, SVR, standalone LSTM forecasting, fixed PRRBM-LSTM fusion, A5 PRRBM-RRCM, and A6 PRX-Net.

Table 5

Cross-dataset average performance, trainable-parameter count, and latency across TEN, CENT, and SE. Latency is measured per forecast window.

Model	RMSE ↓	MAE ↓	MAPE (%) ↓	R^2 ↑	NRMSE (%) ↓	NMAE (%) ↓	Trainable params ↓	Latency (ms) ↓
PRX-Net	740.82	528.89	2.02	0.98	2.90	2.09	73.74K	0.45
iTransformer [22]	759.72	555.09	2.09	0.98	3.00	2.20	211.72K	0.04
PatchTST [21]	838.89	578.54	2.21	0.97	3.33	2.31	59.33K	0.59
edRVFL [24]	850.65	599.05	2.29	0.97	3.38	2.40	0	0.26
LSTM [35]	1115.81	864.97	3.27	0.96	4.45	3.46	469.00K	0.54
TCN [20]	1380.00	1019.14	3.72	0.93	5.31	3.94	49.41K	0.08

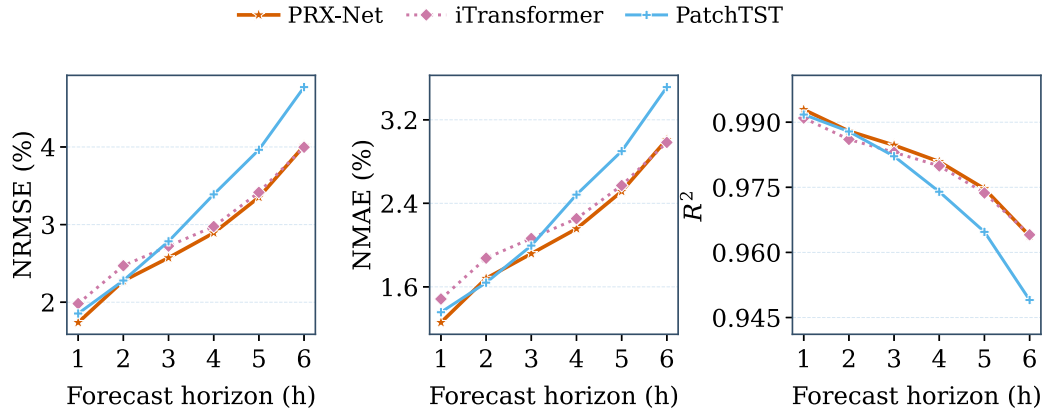


Fig. 4. Cross-dataset horizon-wise NRMSE, NMAE, and R^2 comparison of the top three average-ranked models.

4.2. Comparison with representative forecasting models

This section compares PRX-Net against representative randomized, recurrent, convolutional, and Transformer-style forecasters. edRVFL [24] is the randomized backbone family PRX-Net builds on. LSTM [35] and TCN [20] test recurrent and convolutional sequence learning. iTransformer [22] and PatchTST [21] test recent Transformer-style forecasting. All methods use the same datasets, input and prediction horizons, and chronological split. Table 5 reports the cross-dataset average, where PRX-Net again ranks first across all six metrics. Compared with iTransformer, PRX-Net reduces average RMSE by 2.49%, MAE by 4.72%, and MAPE by 3.48%. Compared with PatchTST, PRX-Net reduces RMSE by 11.69%, MAE by 8.58%, and MAPE by 8.55%. Compared with edRVFL, the reductions reach 12.91% in RMSE, 11.71% in MAE, and 12.08% in MAPE. These results indicate that PRX-Net improves the randomized backbone while remaining competitive with recent Transformer-style forecasters. Table 5 also reports trainable-parameter count and latency. PRX-Net occupies a favorable accuracy–efficiency trade-off point, giving the lowest NMAE with 73.74K trainable parameters, fewer than iTransformer and LSTM, at a latency of 0.45 ms per forecast window. The zero trainable-parameter entry for edRVFL reflects its closed-form ridge-regression solution rather than backpropagation-trained weights, and its lower trainable burden comes with higher NMAE.

Per-region comparisons with representative baselines follow the same pattern as the cross-dataset average. PRX-Net gives the best result across all metrics on Tennessee (RMSE = 517.21, MAPE = 2.07%), Central (RMSE = 947.85, MAPE = 1.89%), and Southeast (RMSE = 757.40, MAPE = 2.08%). The margin over iTransformer is largest on Tennessee (7.87% RMSE reduction) and smallest on Southeast (0.22%), where the two models remain close. Full per-region tables and horizon-wise figures for Tennessee, Central, and Southeast are reported in Tables S8–S10 and Figures S4–S6 of the Supplementary Material. Fig. 4 shows PRX-Net maintaining the lowest average NRMSE and NMAE and the highest R^2 across the horizon, with the gap widening at later steps, indicating that the prototype-routed backbone and recurrent residual correction control error accumulation over the six-hour window.

4.3. Statistical significance analysis

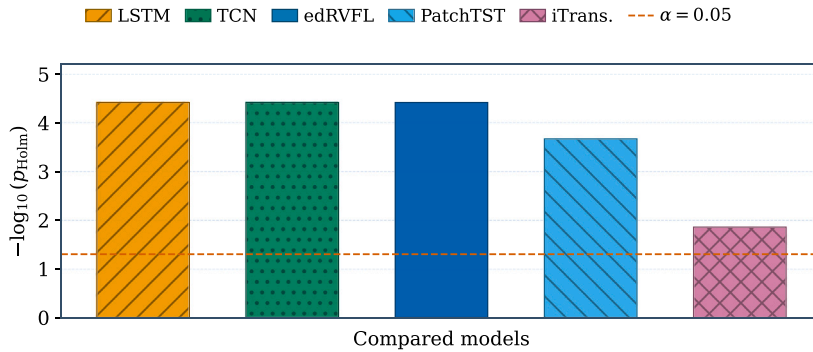
The average and horizon-wise results show that PRX-Net gives the best accuracy across the three-dataset benchmark. The Wilcoxon signed-rank test with Holm correction is applied to test whether these differences persist across paired evaluation units. The paired units match each compared model against PRX-Net over the same dataset and forecast horizon, giving 18 paired comparisons for each baseline. Positive Δ RMSE and Δ MAE values indicate that the compared model has larger error than PRX-Net.

Table 6 reports the corrected significance results for the representative forecasting models. PRX-Net reduces paired RMSE relative to edRVFL, LSTM, TCN, iTransformer, and PatchTST by 97.74, 396.34, 604.41, 24.99, and 84.76, respectively. The Holm-corrected RMSE p -values remain below 0.05 for every comparison, including iTransformer with $p_{\text{Holm}} = 0.0139$ and PatchTST with $p_{\text{Holm}} = 2.14 \times 10^{-4}$. The corrected MAE and R^2 tests show the same trend, which supports the conclusion that PRX-Net improves

Table 6

Wilcoxon signed-rank test with Holm correction for representative baselines against PRX-Net.

Model	Pairs	Δ RMSE	p_{Holm} RMSE	Δ MAE	p_{Holm} MAE	p_{Holm} R^2
LSTM [35]	18	396.34	3.81e-05	336.08	3.81e-05	3.81e-05
PatchTST [21]	18	84.76	0.0002	49.65	0.0021	0.0002
TCN [20]	18	604.41	3.81e-05	490.25	3.81e-05	3.81e-05
edRVFL [24]	18	97.74	3.81e-05	70.16	3.81e-05	3.81e-05
iTransformer [22]	18	24.99	0.0139	26.19	0.0104	0.0208

**Fig. 5.** Wilcoxon signed-rank test with Holm correction for RMSE differences between PRX-Net and the representative forecasting baselines.

accuracy consistently across datasets and horizons. Fig. 5 visualizes the corresponding Holm-corrected RMSE significance results for the representative forecasting baselines.

The corresponding ablation significance results, showing that A6 PRX-Net gives a statistically consistent improvement over every ablation variant (corrected RMSE $p = 2.67 \times 10^{-5}$ throughout), are reported in Table S11 and Figure S7 of the Supplementary Material.

Together, these results show a consistent pattern across all three analyses: the ablation study supports the contribution of each PRX-Net component, and the model comparison and Wilcoxon–Holm analysis confirm that the resulting accuracy gains hold across datasets and forecast horizons.

5. Conclusion

This study proposed PRX-Net, a Prototype-Routed eXponential-Gated Residual Network for multi-horizon regional electricity-load forecasting. The model combined prototype-routed randomized forecasting, recurrent residual correction, and causal horizon-wise error gating to address efficient learning, regime-aware prediction, and reliable residual use in regional load forecasting. Global and local enhanced edRVFL experts captured broad and regime-specific load-weather behavior, while the recurrent module estimated the remaining temporal residual and the exponential previous-error gate regulated its contribution using matured error evidence. Experiments on the TEN, CENT, and SE datasets showed that PRX-Net achieved the best cross-dataset average performance, with RMSE = 740.82, MAE = 528.89, MAPE = 2.02%, $R^2 = 0.9808$, NRMSE = 2.90%, and NMAE = 2.09%. PRX-Net reduced average RMSE by 12.91% compared with edRVFL, 2.49% compared with iTransformer, and 11.69% compared with PatchTST, and the Wilcoxon–Holm analysis supported that these gains were consistent across paired dataset-horizon evaluations. Future work will extend PRX-Net to spatially distributed weather inputs, larger multi-region benchmarks, probabilistic forecasting, adaptive prototype updating, longer forecast horizons, and downstream dispatch or demand-response studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compeleceng.2026.111557>.

Data availability

The raw electricity-load data used in this study are publicly available from the U.S. Energy Information Administration (EIA) Open Data platform. The meteorological variables are publicly available from the National Renewable Energy Laboratory (NREL) National Solar Radiation Database (NSRDB). The processed datasets, forecasting splits, and implementation files can be made available by the corresponding author upon reasonable request.

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